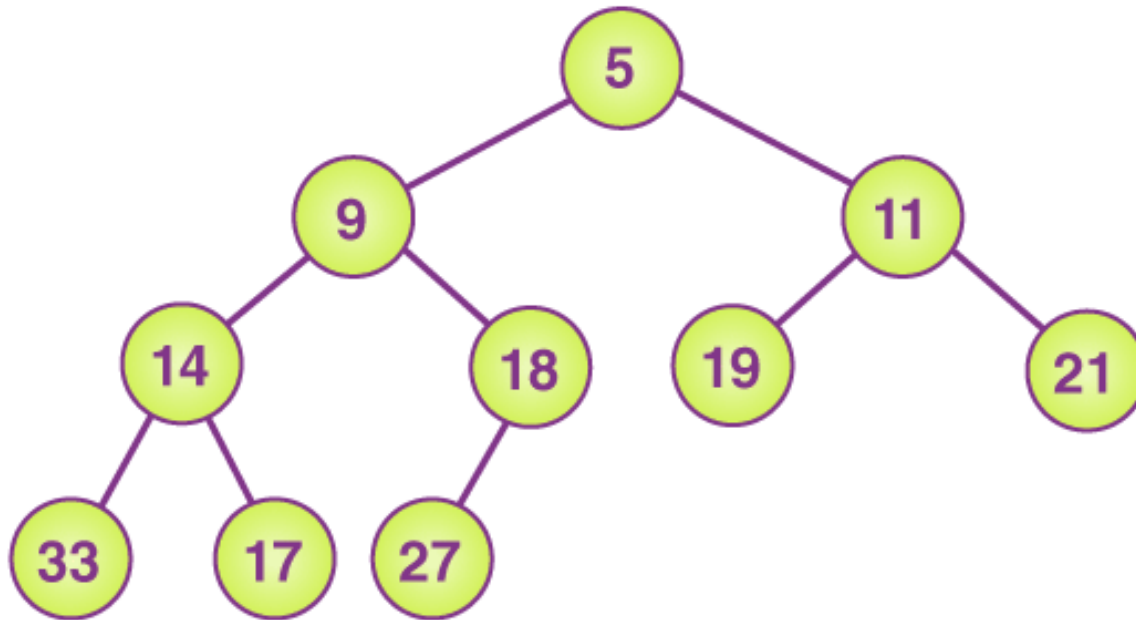
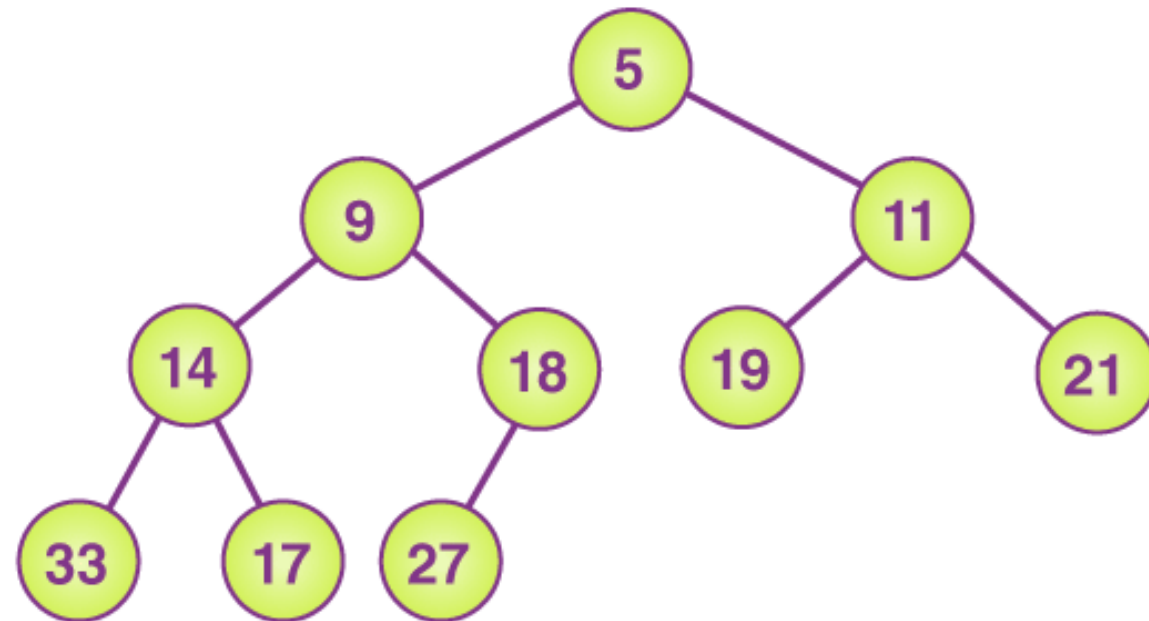


1. Array Representation of Heap

- Please show the array representation of the following min-heap.

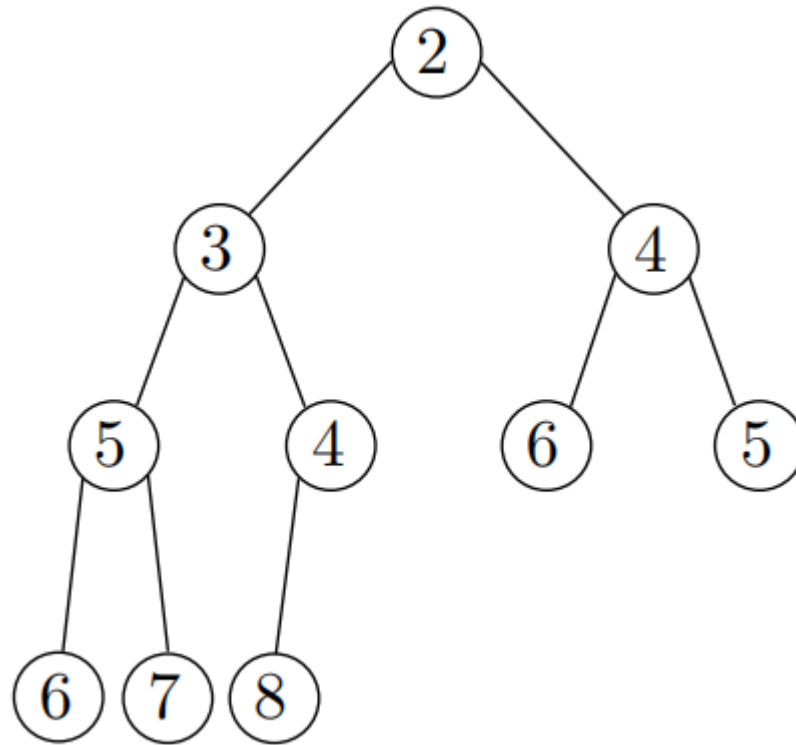




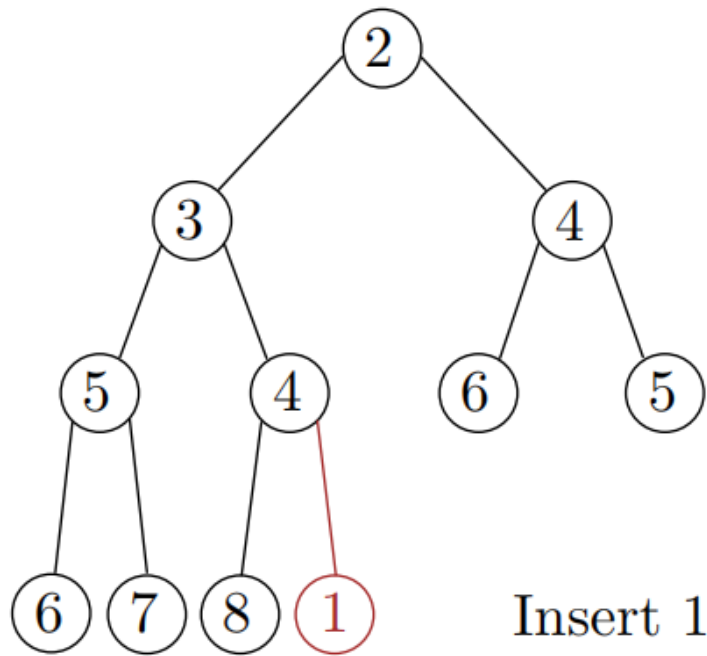
0	5	9	11	14	18	19	21	33	17	27	
0	1	2	3	4	5	6	7	8	9	10	11

2. Insertion Operation of Heap

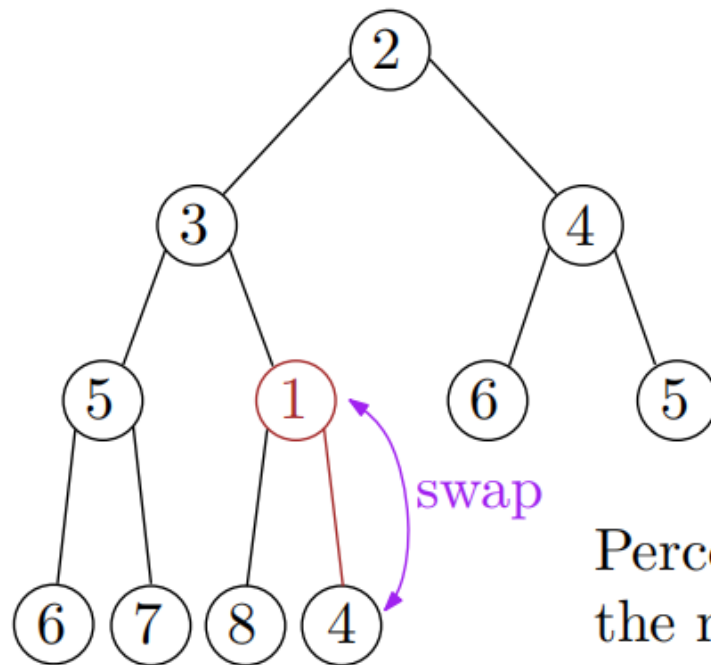
- Please insert 1 into the following min-heap



General strategy is percolate up (or bubble up): if the parent of the element is larger than the element, then interchange the parent with child.

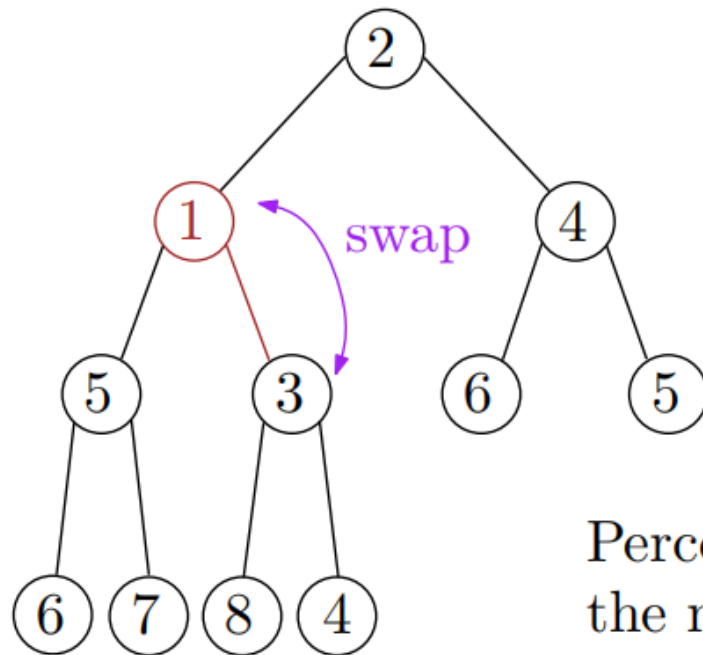


General strategy is percolate up (or bubble up): if the parent of the element is larger than the element, then interchange the parent with child.



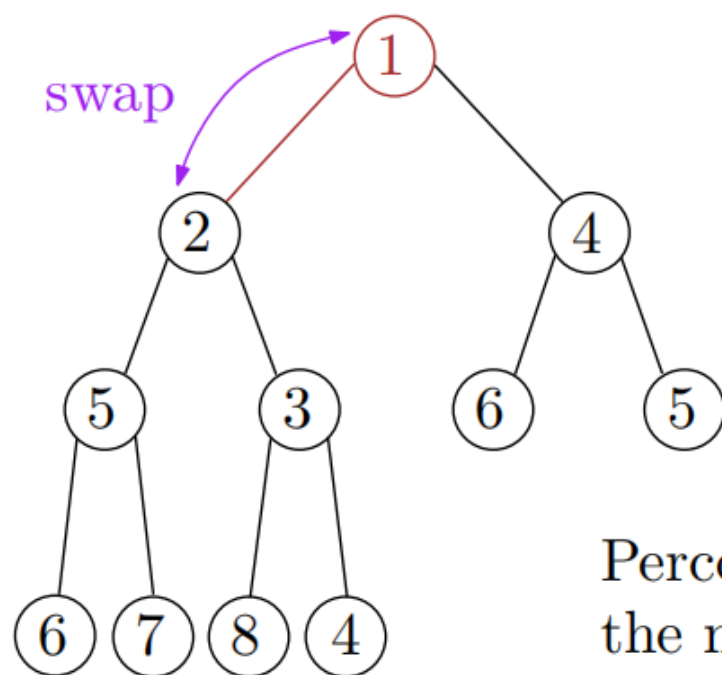
Percolate up to maintain the min-heap property

General strategy is percolate up (or bubble up): if the parent of the element is larger than the element, then interchange the parent with child.



Percolate up to maintain the min-heap property

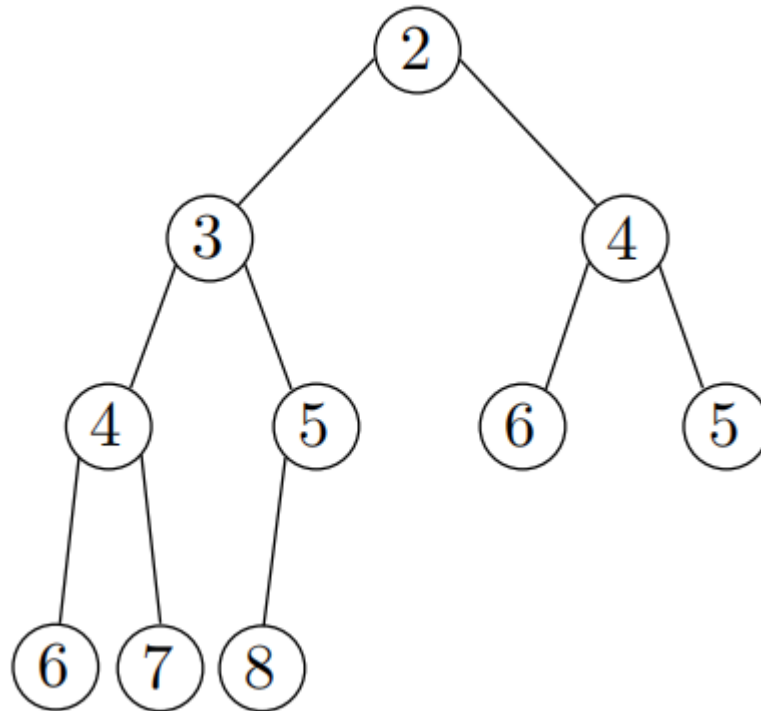
General strategy is percolate up (or bubble up): if the parent of the element is larger than the element, then interchange the parent with child.



Percolate up to maintain the min-heap property

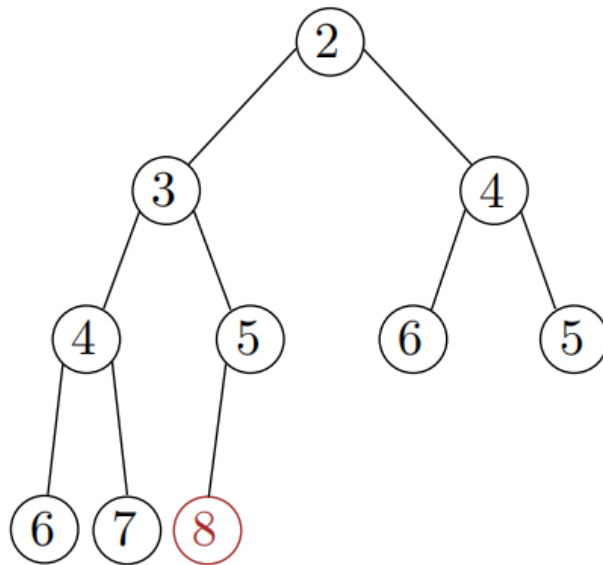
3. Deletion Operation of Heap

- Please show the operation of delete-min (i.e., delete 2) in the following min-heap.



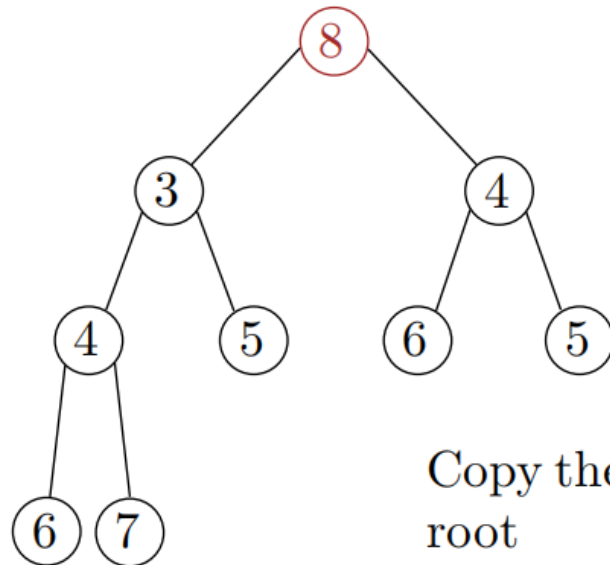
Copy the last element to the root (i.e., overwrite the minimum element stored there)

Restore the min-heap property by percolate down (or bubble down): if the element is larger than either of its children, then interchange it with the smaller of its children.



Copy the last element to the root (i.e., overwrite the minimum element stored there)

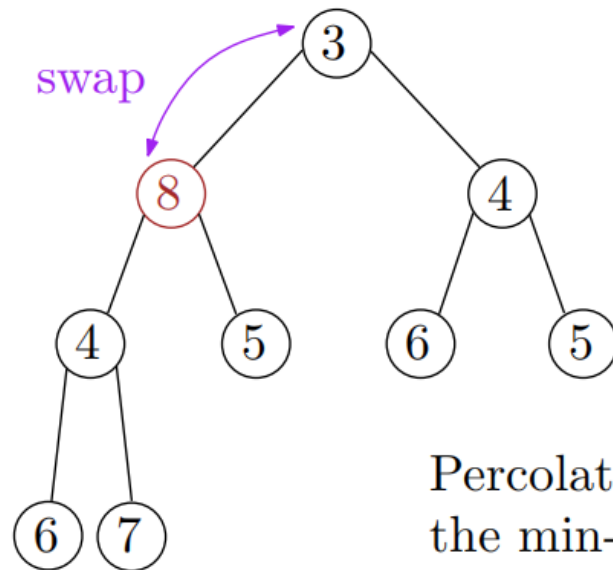
Restore the min-heap property by percolate down (or bubble down): if the element is larger than either of its children, then interchange it with the smaller of its children.



Copy the last element to the root

Copy the last element to the root (i.e., overwrite the minimum element stored there)

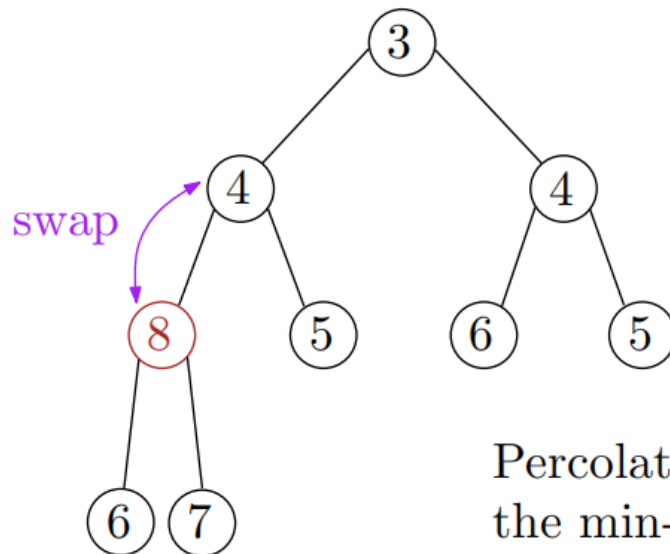
Restore the min-heap property by percolate down (or bubble down): if the element is larger than either of its children, then interchange it with the smaller of its children.



Percolate down to maintain the min-heap property

Copy the last element to the root (i.e., overwrite the minimum element stored there)

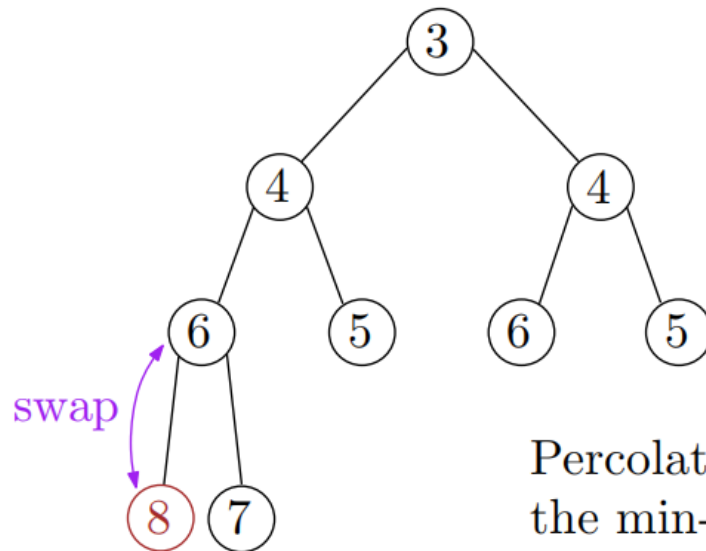
Restore the min-heap property by percolate down (or bubble down): if the element is larger than either of its children, then interchange it with the smaller of its children.



Percolate down to maintain the min-heap property

Copy the last element to the root (i.e., overwrite the minimum element stored there)

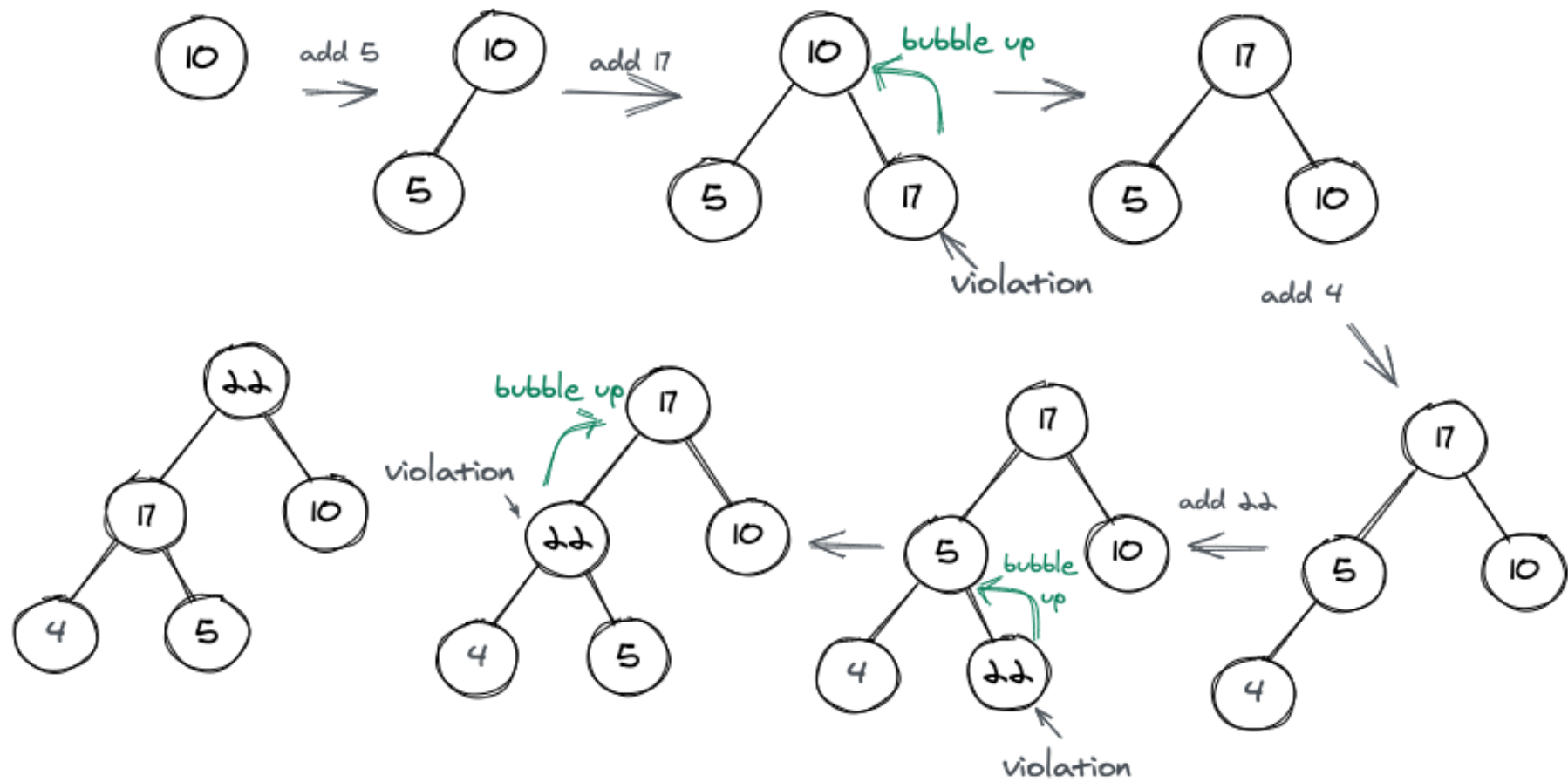
Restore the min-heap property by percolate down (or bubble down): if the element is larger than either of its children, then interchange it with the smaller of its children.



Percolate down to maintain the min-heap property

4. Max-heap Construction

- **Build a max-heap H by inserting the given set of numbers 10,5,17,4,22 into an initially empty max-heap.**



Building a Heap